HDOC-Day 11

- **Find the sum of the squares of its digit using a while loop. E.g Number: 123 Sum=14**
- **Find the sum of cubes of its digits using for loop. E.g Number:123,Sum= 36**
- **Find the difference between the sum of cubes and the sum of squares of its digit using a do while loop. E.g Number: 123, Difference=22**

1. Algorithm

Step 1:

Start the program

Step 2:

Declare variables for the number, choice, digit, sum of squares, sum of cubes and difference.

Step 3:

Input a positive integer from the user.

Step 4:

Display the menu:

1 → Sum of squares

2 → Sum of cubes

3 → Difference between sum of cubes and sum of squares

4 → Exit

Step 5:

Input the user's choice.

Step 6:

Use switch-case according to the choice.

Case 1: Sum of squares

Set sumSquare = 0.

Extract each digit using % 10.

Add the square of the digit to sumSquare.

Remove the last digit using / 10.

Repeat using a while loop.

Display the result

Case 2: Sum of cubes

Set sumCube = 0.

Use a for loop to extract every digit.

Add the cube of each digit to sumCube.

Display the result

Case 3: Difference

Calculate the sum of squares and sum of cubes using a do-while loop.

Calculate:

Difference = Sum of Cubes - Sum of Squares

Display the result.

Case 4:

Exit the program.

Step 7:

Stop

Test Case Table:

Test Case	Input Number	Choice	Expected Output

1	123	1	Sum of squares=14
2	123	2	Sum of cubes=36
3	123	3	Difference=22
4	45	1	Sum of squares= 41
5	45	2	Sum of cubes= 189
6	45	3	Difference= 148
7	7	1	Sum of squares=49
8	7	2	Sum of Cubes= 343
9	7	3	Difference= 294
10	123	4	Program exits

• C Program:

```
#include <stdio.h>


int main()

{

   int num, choice, temp, digit;

   int sumSquare, sumCube, difference;


   printf("Enter a positive integer: ");

   scanf("%d", &num);


   printf("\n----- MENU -----\n");

   printf("1. Sum of squares of digits\n");

   printf("2. Sum of cubes of digits\n");

   printf("3. Difference between sum of cubes and sum of squares\n");

   printf("4. Exit\n");
```

```
printf("Enter your choice: ");
scanf("%d", &choice);

switch(choice)
{
  case 1:
    sumSquare = 0;
    temp = num;

    while(temp > 0)
    {
      digit = temp % 10;
      sumSquare = sumSquare + digit * digit;
      temp = temp / 10;
    }

    printf("Sum of squares = %d\n", sumSquare);
    break;

  case 2:
    sumCube = 0;
    temp = num;

    for(; temp > 0; temp = temp / 10)
    {
      digit = temp % 10;
      sumCube = sumCube + digit * digit * digit;
```

```
  }

  printf("Sum of cubes = %d\n", sumCube);
  break;

case 3:
  sumSquare = 0;
  sumCube = 0;
  temp = num;

  do
  {
    digit = temp % 10;

    sumSquare = sumSquare + digit * digit;
    sumCube = sumCube + digit * digit * digit;

    temp = temp / 10;

  } while(temp > 0);

  difference = sumCube - sumSquare;

  printf("Sum of squares = %d\n", sumSquare);
  printf("Sum of cubes = %d\n", sumCube);
  printf("Difference = %d\n", difference);

  break;
```

```
    case 4:
      printf("Exiting the program...\n");
      break;

    default:
      printf("Invalid choice!\n");
  }

  return 0;
}
```

Compilation and Execution:

Test Case1:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter a positive integer:123

───── MENU ─────
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
4. Exit
Enter your choice: 1
Sum of squares = 14
```

Test Case 2:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter a positive integer:123

───── MENU ─────
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
4. Exit
Enter your choice: 2
Sum of cubes = 36
```

Test Case 3:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter a positive integer:123

───── MENU ─────
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
4. Exit
Enter your choice: 3
Sum of squares = 14
Sum of cubes = 36
Difference = 22
```

Test Case 4:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter a positive integer:45

───── MENU ─────
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
4. Exit
Enter your choice: 1
Sum of squares = 41
```

Test Case 5:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter a positive integer:45

───── MENU ─────
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
4. Exit
Enter your choice: 2
Sum of cubes = 189
```

Test Case 6:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter a positive integer:45

───── MENU ─────
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
4. Exit
Enter your choice: 3
Sum of squares = 41
Sum of cubes = 189
Difference = 148
```

Test Case 7:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter a positive integer:7

───── MENU ─────
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
4. Exit
Enter your choice: 1
Sum of squares = 49
```

Test Case 8:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter a positive integer:7

───── MENU ─────
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
4. Exit
Enter your choice: 2
Sum of cubes = 343
```

Test Case 9:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter a positive integer:7

───── MENU ─────
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
4. Exit
Enter your choice: 3
Sum of squares = 49
Sum of cubes = 343
Difference = 294
```

Test Case 10:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter a positive integer:123

────── MENU ──────
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
4. Exit
Enter your choice: 4
Existing the program...
```